

1. Set up an integral to find the area of the region bounded by $y = 2x^2$ and $y = x^3$.
2. Set up an integral to find the volume of the solid generated when the region bounded by $y = 2x^2$ and $y = x^3$ is
 - (a) Rotated about the x -axis using shells
 - (b) Rotated about the x -axis using washers
 - (c) Rotated about the y -axis using shells
 - (d) Rotated about the y -axis using washers

(e) Rotated about the line $x = 3$

(f) Rotated about the line $y = -2$

(g) Rotated about the line $y = 11$

(h) Rotated about the line $x = -7$

3. Suppose a solid has the region bounded by $y = 2x^2$ and $y = x^3$ as a base. Find the volume if the cross-sections are

(a) Circles

(b) Semi-circles

(c) Rectangles in which the height is 3 times the base